MPLS Layer 3 VPN Project Lab

Hands-on lab project | OSPF, LDP, MP-BGP VPNv4, VRF and Route Targets | Cisco IOS

1. Project Overview

I designed and configured a service-provider MPLS backbone that connects each customer's remote sites over a shared core, without their traffic or IP space ever mixing. The goal was to understand how an MPLS Layer 3 VPN works by building every layer myself, from basic IP connectivity up to end-to-end customer reachability.

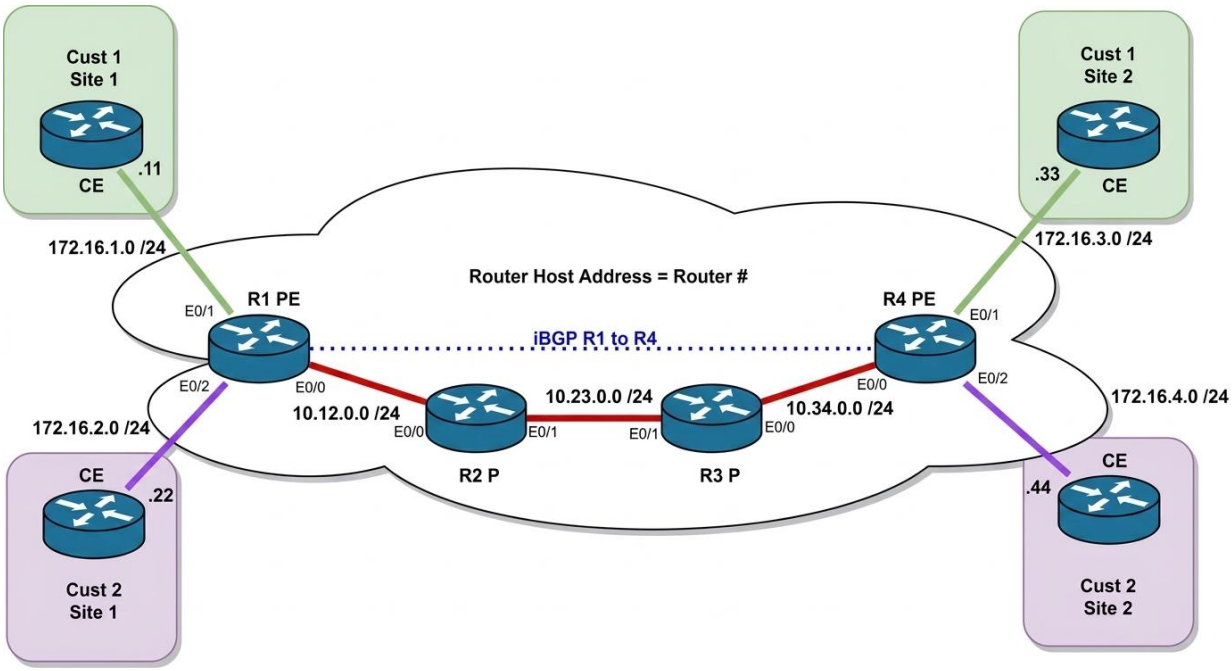
Item	Detail
Provider routers	4 (R1 and R4 are PE routers; R2 and R3 are P routers)
Customers	2 (Customer 1 and Customer 2), 2 sites each, 4 sites in total
BGP AS number	6783 (iBGP between the PEs)
Underlay IGP	OSPF process 1, area 0
Label distribution	LDP
Customer routing	OSPF between PE and CE (process 10 for cust1, 20 for cust2)

Build roadmap:

1. Core IP addressing and OSPF underlay
2. MPLS and LDP label distribution
3. MP-BGP VPNv4 between the PE routers
4. VRFs, Route Distinguishers and Route Targets
5. PE-CE interface binding and route redistribution
6. CE configuration and verification.

2. Topology and Addressing

R1 and R4 are the provider edge (PE) routers. R2 and R3 are core (P) routers. Customer 1 (VRF cust1) and Customer 2 (VRF cust2) each have one site on each PE.



Loopback0 addresses (/32)

Router	Loopback0
R1	1.1.1.1/32
R2	2.2.2.2/32
R3	3.3.3.3/32
R4	4.4.4.4/32

Link addressing

Link	Subnet	Endpoints (host ID)	VRF
R1 - R2	10.12.0.0/24	R1 E0/0 (.1), R2 E0/0 (.2)	-
R2 - R3	10.23.0.0/24	R2 E1/0 (.2), R3 E1/0 (.3)	-
R3 - R4	10.34.0.0/24	R3 E0/0 (.3), R4 E0/0 (.4)	-
R1 - C1S1	172.16.1.0/24	R1 E0/1 (.1), C1S1 (.11)	cust1
R1 - C2S1	172.16.2.0/24	R1 E0/2 (.1), C2S1 (.22)	cust2
R4 - C1S2	172.16.3.0/24	R4 E0/1 (.4), C1S2 (.33)	cust1
R4 - C2S2	172.16.4.0/24	R4 E0/2 (.4), C2S2 (.44)	cust2

3. Phase 1: Core Addressing and OSPF Underlay

The provider routers need basic IP connectivity so that every Loopback0 can reach every other one. LDP and the BGP sessions both depend on this. OSPF (process 1, area 0) runs on all core interfaces and loopbacks.

R1 (example; the other routers follow the same pattern with their own addresses):

```
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
 ip ospf 1 area 0
!
interface Ethernet0/0
 ip address 10.12.0.1 255.255.255.0
 ip ospf 1 area 0
!
router ospf 1
```

Verify: `show ip ospf neighbor`, then `ping 4.4.4.4 source Loopback0` from R1.

4. Phase 2: Enabling MPLS in the Core

This step turns the routed network into an MPLS backbone. Enable CEF, give each router its own label range, and run LDP on every core-facing interface.

```
ip cef
!
mpls label range 100 199
!
interface Ethernet0/0
 ip ospf 1 area 0
 mpls ip
```

Router	Label range
R1	100 - 199
R2	200 - 299
R3	300 - 399
R4	400 - 499

Why separate label ranges? In `show mpls forwarding-table`, you can tell at a glance which router allocated each label.

Verify: `show mpls ldp neighbor` and `show mpls forwarding-table`.

5. Phase 3: MP-BGP Between the PE Routers

R1 and R4 exchange customer routes in VPNv4 format over iBGP (AS 6783), peering loopback to loopback.

On R1:

```
router bgp 6783
 neighbor 4.4.4.4 remote-as 6783
 neighbor 4.4.4.4 update-source Loopback0
 neighbor 4.4.4.4 next-hop-self
 !
 address-family vpnv4
  neighbor 4.4.4.4 activate
  neighbor 4.4.4.4 send-community extended
 exit-address-family
```

On R4:

```
router bgp 6783
 neighbor 1.1.1.1 remote-as 6783
 neighbor 1.1.1.1 update-source Loopback0
 neighbor 1.1.1.1 next-hop-self
 !
 address-family vpnv4
  neighbor 1.1.1.1 activate
  neighbor 1.1.1.1 send-community extended
 exit-address-family
```

- `update-source Loopback0`: the session survives any single link failure.
- `address-family vpnv4`: carries the RD plus the prefix, so overlapping customer routes stay unique.

- `send-community extended`: required to carry the Route Targets.
- Only the PEs run BGP. The P routers (R2, R3) stay BGP-free.

6. Phase 4: VRFs, Route Distinguishers and Route Targets

Each VRF is a private routing table per customer. Route Targets decide which routes are shared between sites. The import and export values are swapped between R1 and R4 so the remote sites can reach each other.

Router	VRF	RD	RT export	RT import
R1	cust1	11:33	11:11	33:33
R4	cust1	11:33	33:33	11:11
R1	cust2	22:44	22:22	44:44
R4	cust2	22:44	44:44	22:22

On R1:

```
vrf definition cust1
 rd 11:33
 address-family ipv4
  route-target export 11:11
  route-target import 33:33
 exit-address-family
!
vrf definition cust2
 rd 22:44
 address-family ipv4
  route-target export 22:22
  route-target import 44:44
 exit-address-family
```

On R4:

```
vrf definition cust1
 rd 11:33
 address-family ipv4
  route-target export 33:33
  route-target import 11:11
 exit-address-family
!
vrf definition cust2
 rd 22:44
 address-family ipv4
  route-target export 44:44
  route-target import 22:22
 exit-address-family
```

The key idea: what R1 exports, R4 imports, and vice versa. Match them and the sites see each other; mismatch them and they do not.

7. Phase 5: Binding Interfaces and PE-CE Routing

7.1 Bind the customer-facing interfaces to the VRFs

On R1:

```
interface Ethernet0/1
 description PE-R1 to Customer 1 Site 1
 vrf forwarding cust1
 ip address 172.16.1.1 255.255.255.0
!
interface Ethernet0/2
 description PE-R1 to Customer 2 Site 1
 vrf forwarding cust2
 ip address 172.16.2.1 255.255.255.0
```

Order matters: apply `vrf forwarding` before `ip address`. Applying a VRF afterwards erases the IP address.

R4 mirrors this: Ethernet0/1 in cust1 (172.16.3.4) and Ethernet0/2 in cust2 (172.16.4.4).

7.2 PE-CE routing and mutual redistribution

On R1:

```
router ospf 10 vrf cust1
 redistribute bgp 6783 subnets
 network 172.16.1.0 0.0.0.255 area 0
!
router ospf 20 vrf cust2
 redistribute bgp 6783 subnets
 network 172.16.2.0 0.0.0.255 area 0
!
router bgp 6783
 address-family ipv4 vrf cust1
  redistribute ospf 10
  redistribute connected
 exit-address-family
 address-family ipv4 vrf cust2
  redistribute ospf 20
  redistribute connected
 exit-address-family
```

- **OSPF into BGP:** routes learned from the CE are injected into the VRF's BGP table.
- **BGP into OSPF:** routes from the remote PE are handed to the local CE.
- **redistribute connected:** advertises the PE-CE subnet itself.

R4 uses the same structure with OSPF processes 10 and 20.

8. Phase 6: Customer Edge (CE) Configuration

Each CE enables OSPF on its PE-facing interface. The same template is used on all four CE routers. The CE routers know nothing about MPLS, VRFs or BGP; they just run plain OSPF toward the PE.

```
interface Ethernet0/1
 ip address 172.16.1.11 255.255.255.0
 ip ospf 1 area 0
!
router ospf 1
```

CE	Customer / site	IP address
C1S1	Customer 1, Site 1	172.16.1.11
C2S1	Customer 2, Site 1	172.16.2.22
C1S2	Customer 1, Site 2	172.16.3.33
C2S2	Customer 2, Site 2	172.16.4.44

9. Verification

9.1 PE VRF routing table (R4)

The remote customer prefix is learned over MP-iBGP from the opposite PE (1.1.1.1). The BGP distance of 200 confirms it is an iBGP route.

```
R4#show ip route vrf cust1

      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
B        172.16.1.0/24 [200/0] via 1.1.1.1, 00:02:17
C        172.16.3.0/24 is directly connected, Ethernet0/1
L        172.16.3.4/32 is directly connected, Ethernet0/1
```

9.2 Remote customer site routing table (C1S2)

C1S2 sees its own connected subnet and learns Customer 1 Site 1's prefix (172.16.1.0/24) over OSPF from the PE.

```
C1S2#show ip route

      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O IA     172.16.1.0/24 [110/11] via 172.16.3.4
C        172.16.3.0/24 is directly connected, Ethernet0/1
L        172.16.3.33/32 is directly connected, Ethernet0/1
```

9.3 End-to-end ping (C1S1 to C1S2)

```
C1S1#ping 172.16.3.33
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.33, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

What a successful ping proves:

- **OSPF underlay:** loopbacks are reachable across the core.
- **LDP:** transport labels switch packets hop by hop.
- **MP-iBGP VPNv4:** customer routes are exchanged between the PEs.

10. Break It to Learn It

Two deliberate failures that show how the layers depend on each other.

Experiment	Action	Result	Lesson
Break MPLS	Run `no mpls ip` on the R2 or R3 core interfaces.	BGP and VRF tables still look healthy, but customer pings fail because the core can no longer label-switch VPN traffic.	A healthy control plane does not guarantee a working data plane.
Break the Route Targets	Change R4's import RT to a value nothing exports.	R4's cust1 VRF stops importing Site 1's prefix, so Customer 1 Site 2 loses visibility of Site 1.	Route Targets are what enforce multi-tenant segregation.

11. Key Takeaways

- **BGP decides where, labels decide how.** MPLS separates the control plane from the data plane.
- **Keep the core simple.** Only the PEs carry customer routes; the P routers just switch labels.
- **RD makes routes unique, RT controls who sees them.** Two different jobs.
- **Customers stay simple.** The CE runs an ordinary IGP while the provider hides the complexity.

Skills practiced: MPLS, LDP, MP-BGP, VPNv4, VRF, OSPF, Route Targets, Cisco IOS, network design, troubleshooting.